

INFOSYS SPRINGBOARD

AirFly Insights

Data Visualization and Analysis of Airline Operations



Domain – Data Visualization
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WHY THIS JOURNEY ?



The global airline network runs millions of flights every year, yet passengers and airlines still struggle with delays, cancellations, and unpredictable travel experiences. To operate efficiently and keep customers satisfied, airlines need a clear view of when and where disruptions happen, and why.

This project set out to turn a massive flight dataset into a navigational map that reveals operational bottlenecks, risky time periods, and high-performing routes, so decision-makers can move from guesswork to data-driven planning.

MEET OUR TRAVEL LOG

- Used Kaggle Airlines Flights data as a large-scale **flight log** capturing **operational details** for each journey.
- Merged three core files: **airlines.csv**, **airports.csv**, and **flights.csv** to build a single **enriched dataset**.
- **airlines.csv** added **airline information** through columns like **IATA_CODE** and **AIRLINE**.
- **airports.csv** contributed **airport context** such as **code**, **city**, **state**, **country**, **latitude**, and **longitude**.
- **flights.csv** described each flight with **date**, **airline**, **flight number**, **origin**, **destination**, **scheduled** and **actual times**.
- Included disruption details like **arrival delay**, **diverted flag**, **cancelled flag**, and **cancellation reason**.
- Also captured granular delay types: **air system**, **security**, **airline**, **late aircraft**, and **weather delays**.
- After combining these sources, each row became a complete story of one flight: **who flew it, from where to where, when it flew, and how it was disrupted**.



PREPARING THE AIRCRAFT

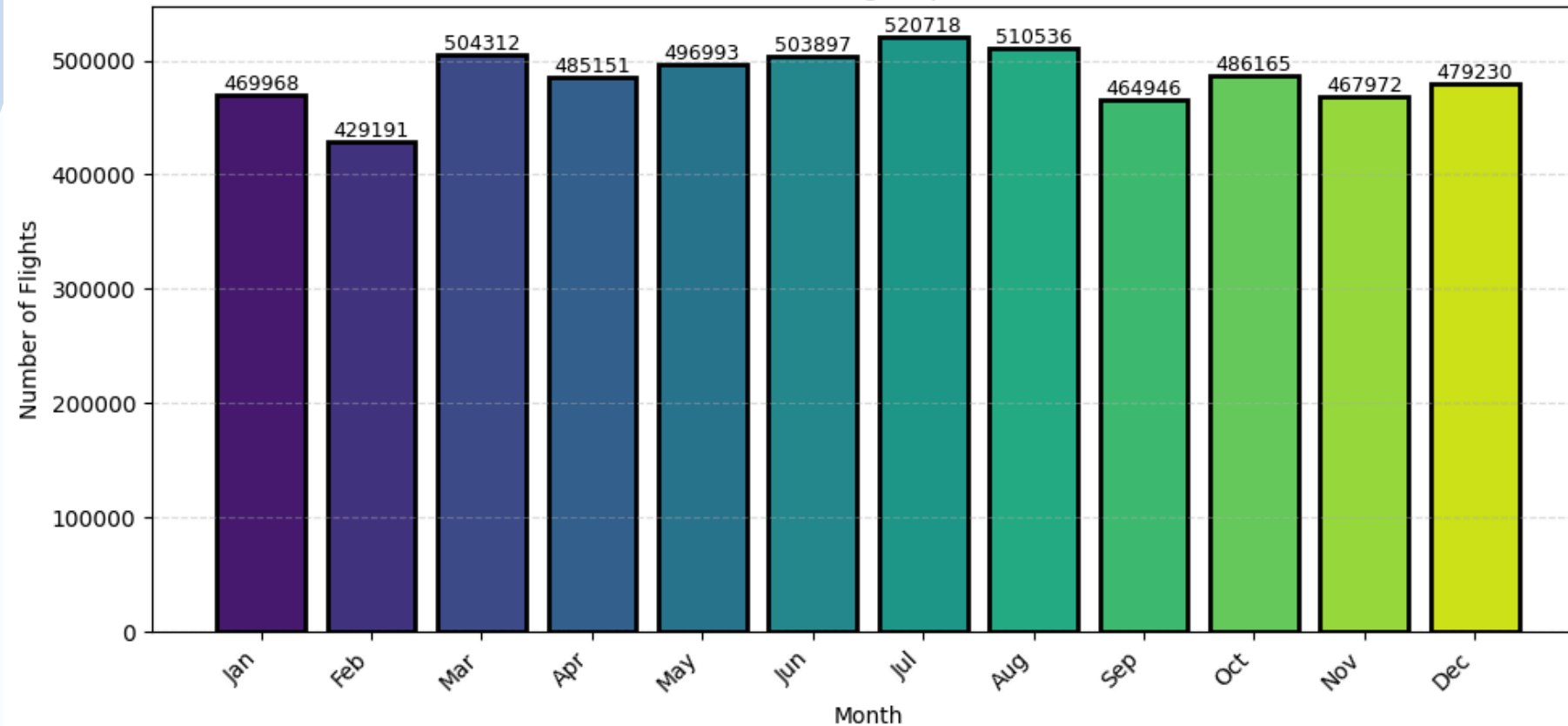
Before take-off, the raw data needed a maintenance check: **missing values** were handled, inconsistent codes were fixed, and impossible values (like negative flight times) were removed. **New features** were engineered to make the story clearer, such as **flags for peak seasons and holidays**, **total delay minutes**, and categorical tags for delay causes and cancellation types. This transformed a noisy logbook into a clean, structured flight diary where each record could be trusted to reflect real operational behavior.



UNIVARIATE AND BIVARIATE



Numbers of Flights per Month



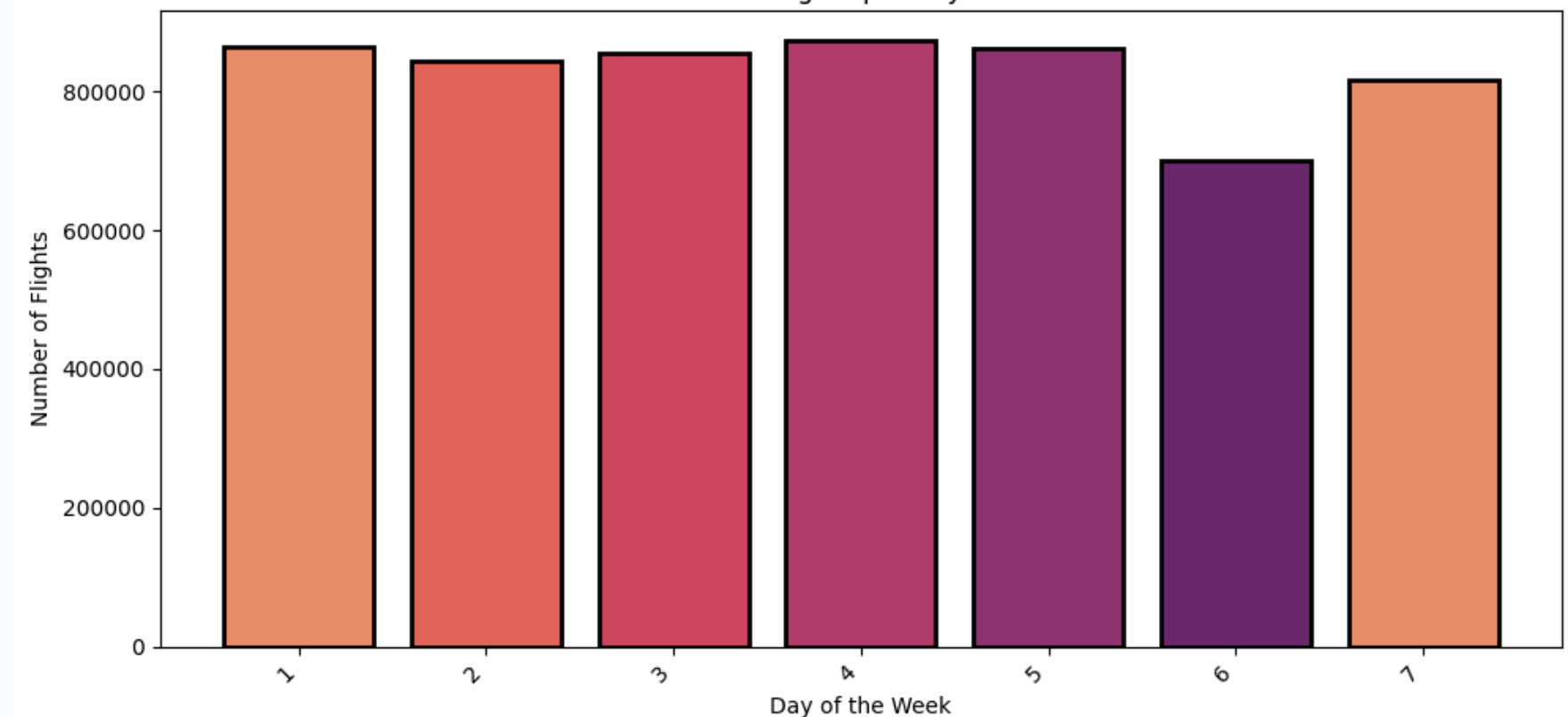
Number of flights per month

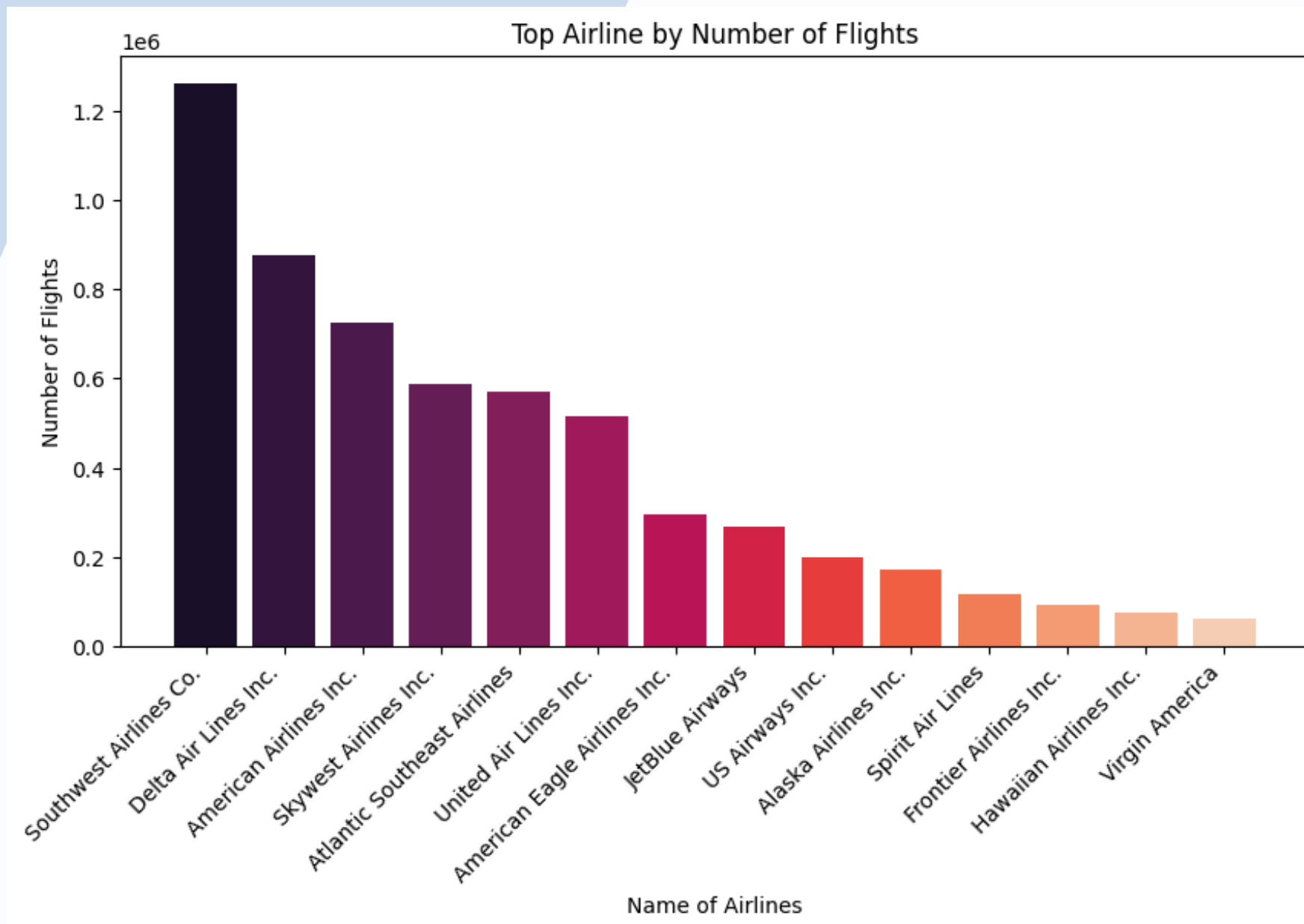
- Flights per month stay almost constant, with only slight mid-year increases in volume..
- Overall, July month took the highest number of flights.

Number of flights per day in a week

- Flights per weekday are higher than on weekends, showing heavier traffic on working days.

Number of flights per day in a week



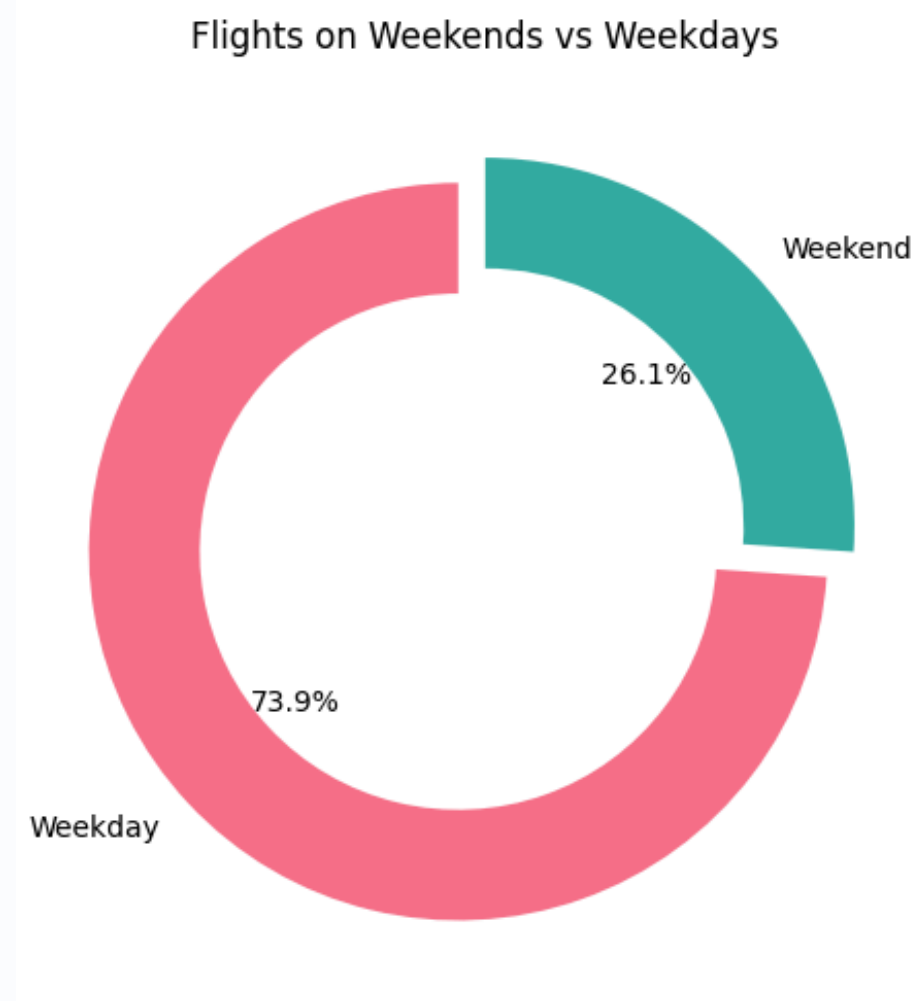


Number of flights per airline

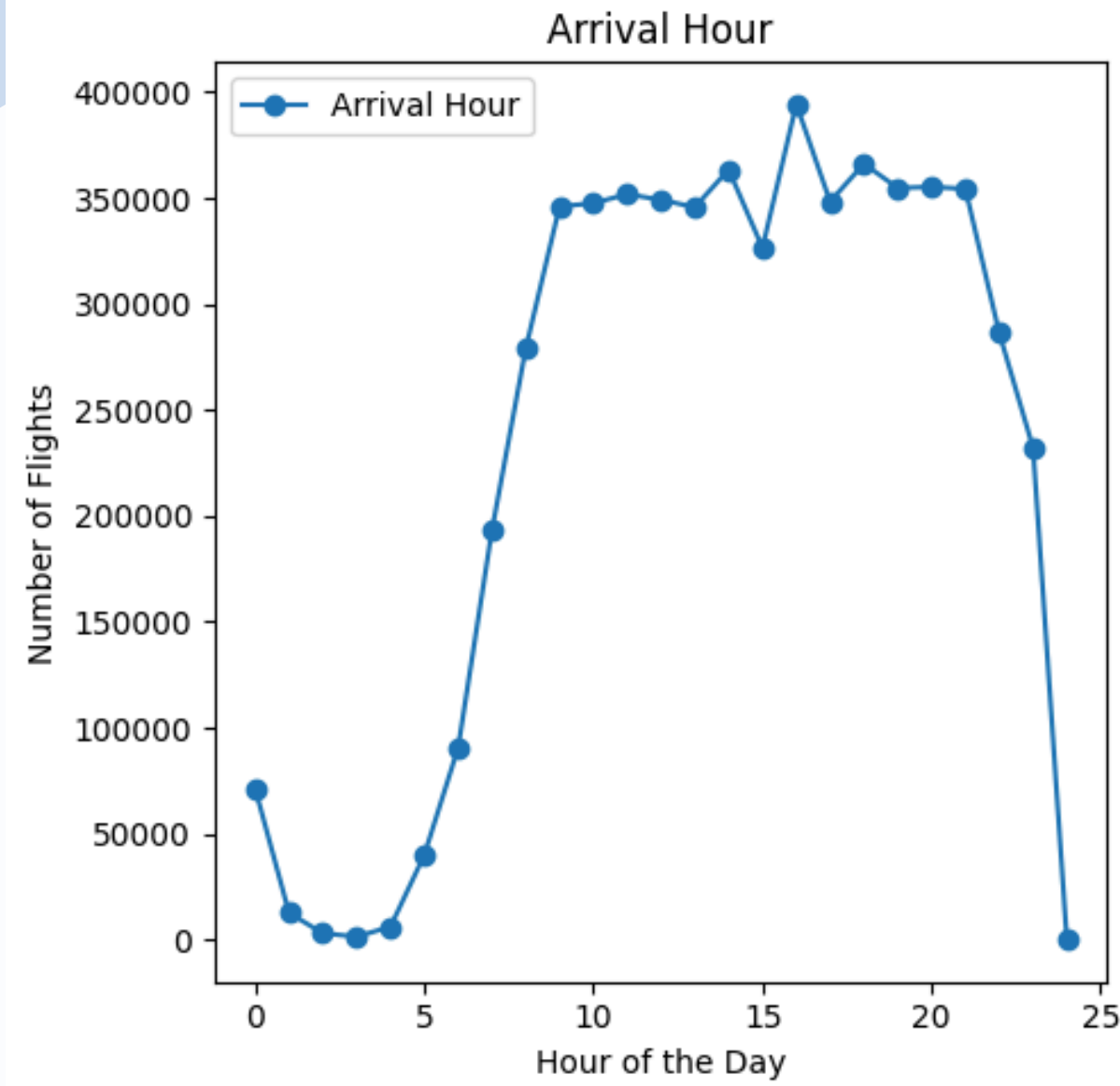
- **Southwest Airlines** operated the **highest** number of flights in the dataset.
- **Virgin America** operated the **lowest** number of flights among the airlines analyzed

Weekend Vs Weekdays flights

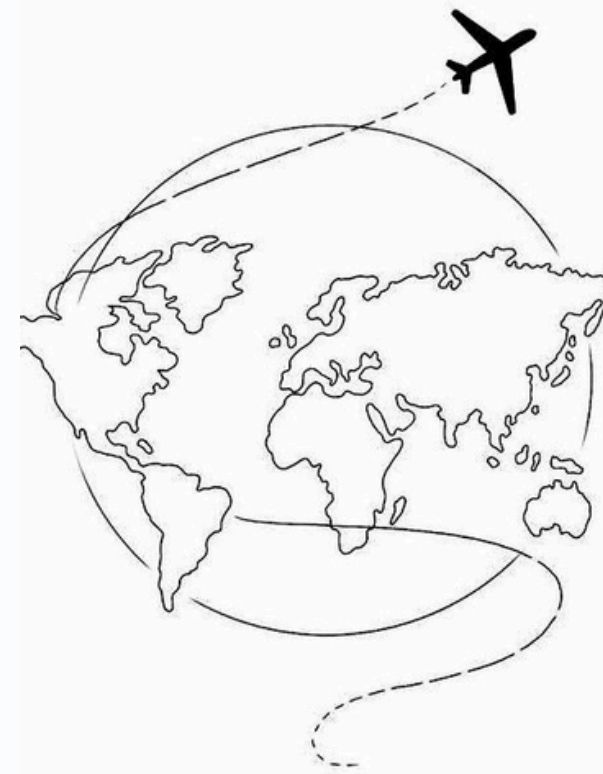
- **Weekday flights** are **higher** than weekend flights, indicating **heavier traffic** and operational load on working days.



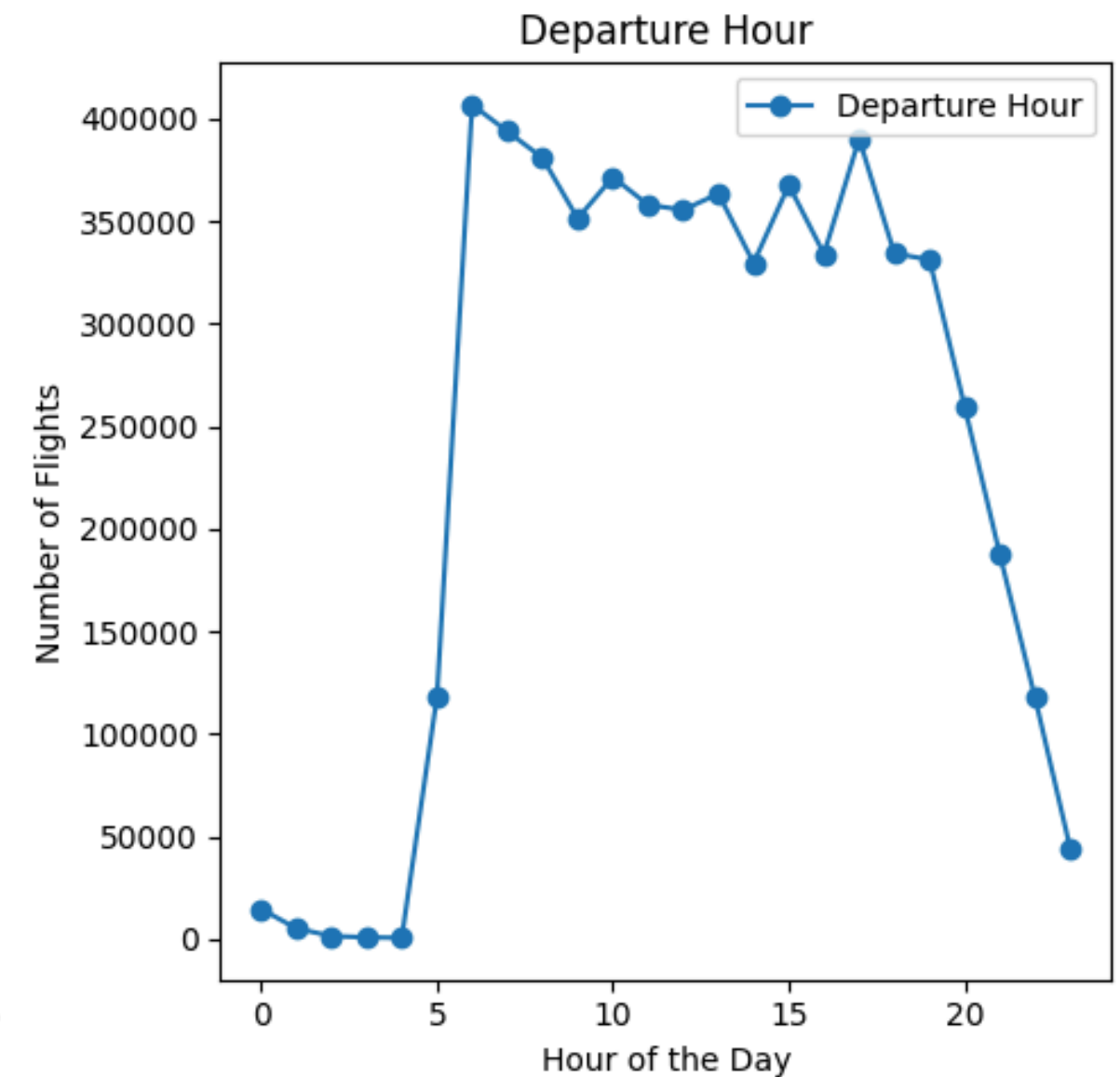
Flights By Arrival & Departure Hour

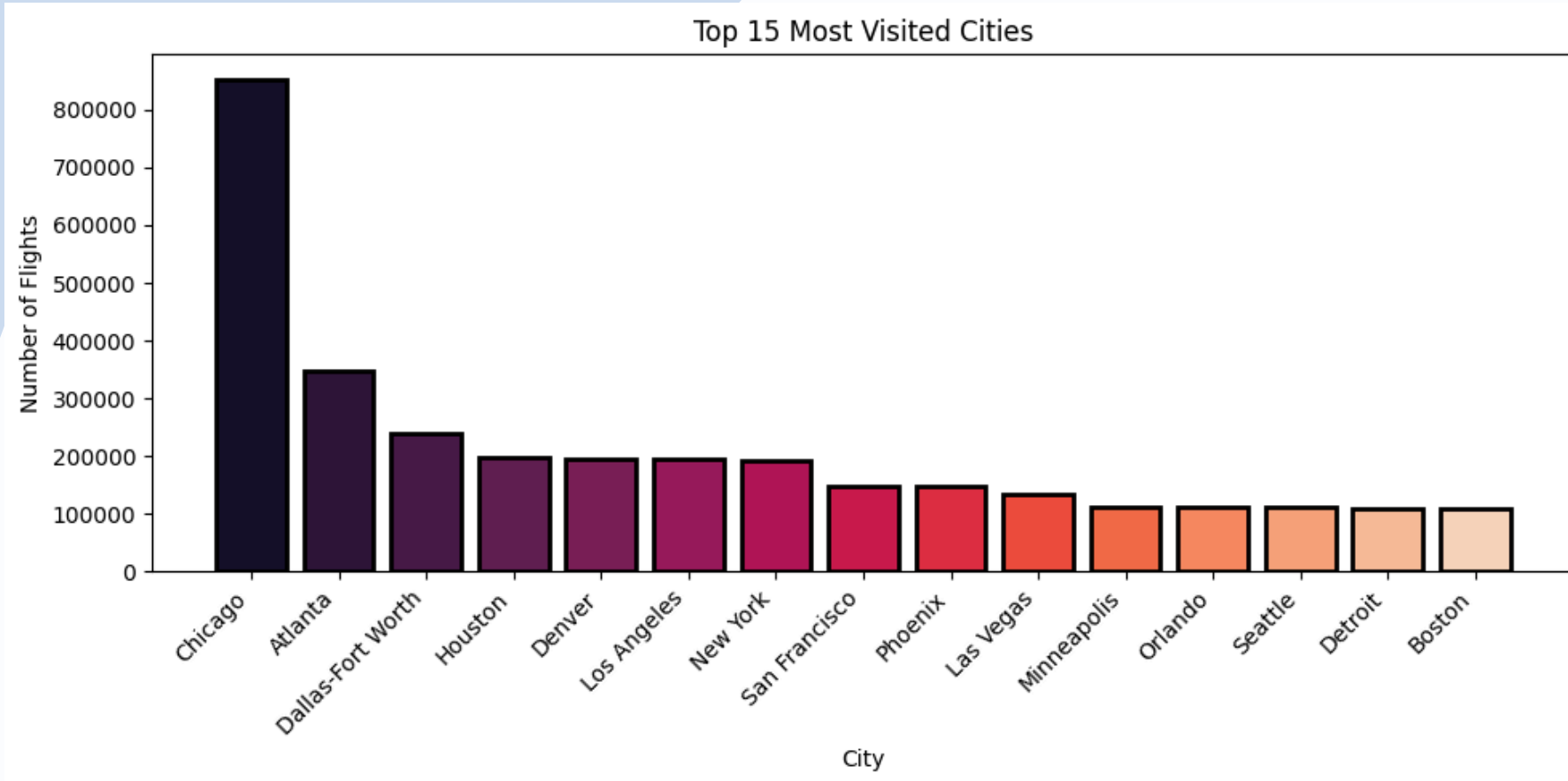


- **Peak hours:** Flights are **highest** during the daytime (**around 9 AM to 8 PM**) and lowest at night.



- **Same trend:** Arrivals and departures follow a **similar pattern** throughout the day.





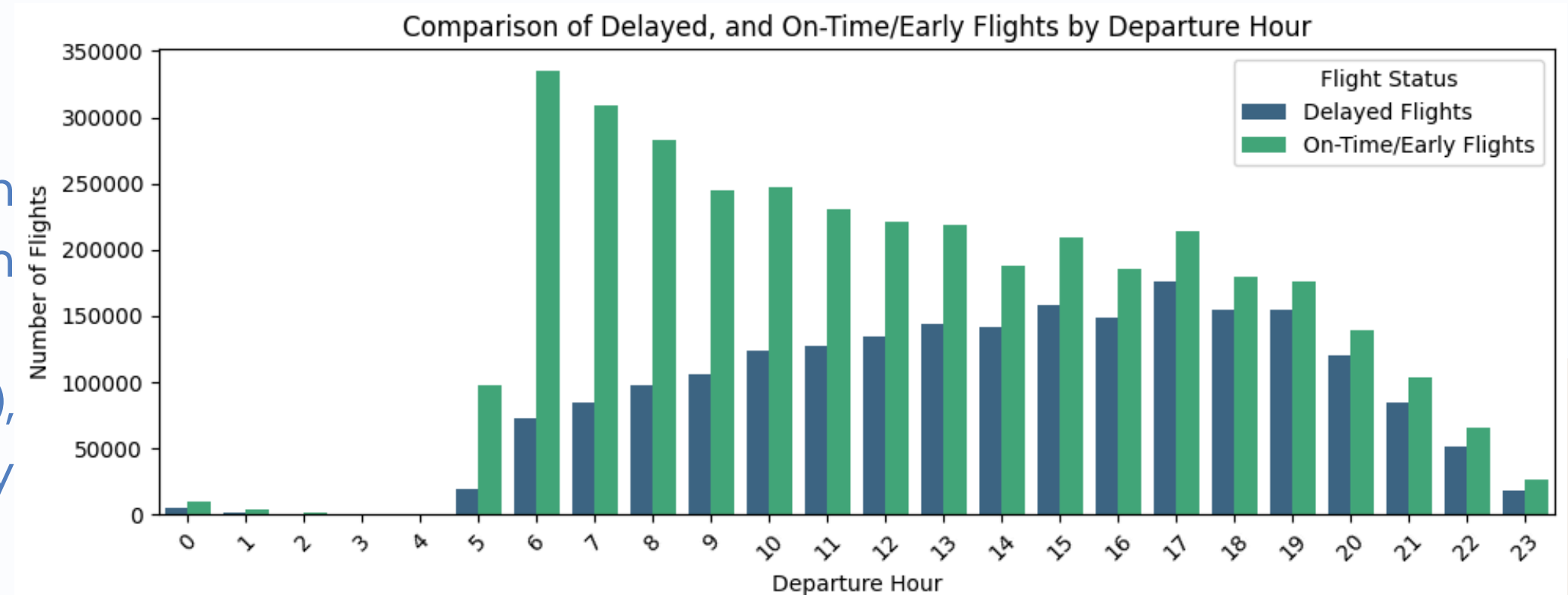
Top 15 most visited cities

- **Chicago** records the **highest number of flights** in the dataset, indicating its role as a major operational hub.



Delayed Flights Vs Flights On Time

- **Early-morning flights (5–9 AM)** are mostly on time, with **delays** beginning to rise from **mid-morning** onward.
- **Delay** levels peak during the **evening (4–8 PM)**, and although **late-night flights are fewer**, they still experience noticeable delays.

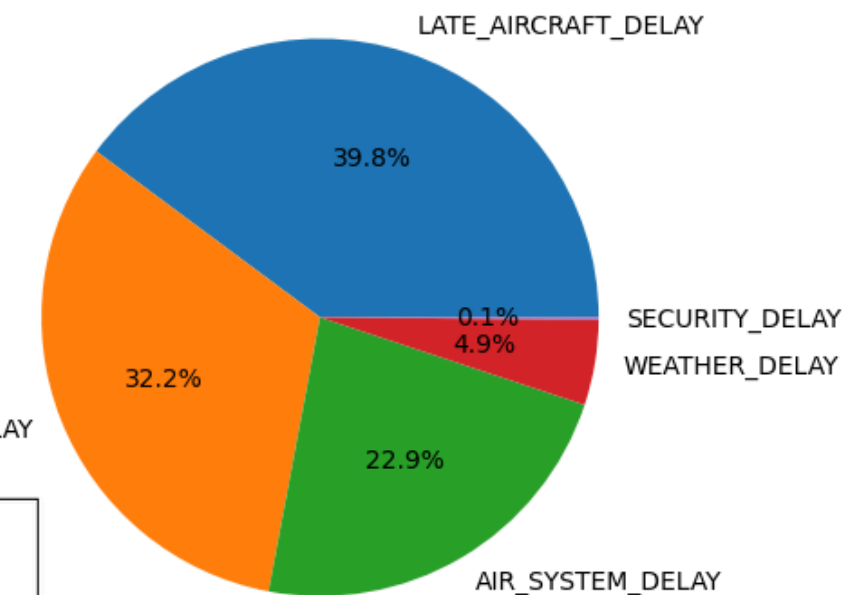




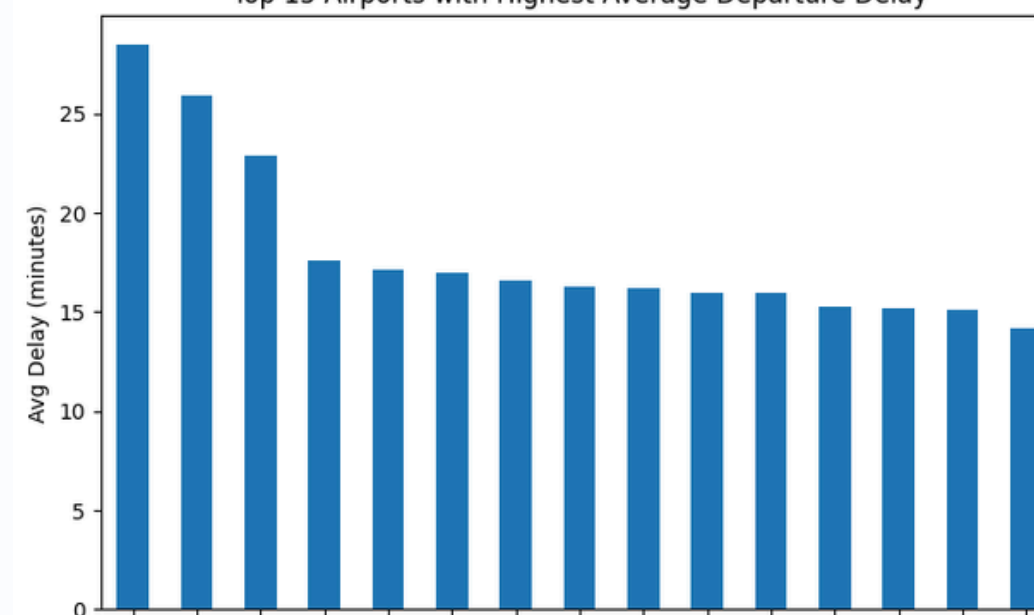
DELAY ANALYSIS - AIRLINE & WEATHER

- “**Late aircraft delay**” is the main delay cause for most airlines, while “**weather and security**” delays are **minimal**, with weather causing the fewest delays.
- “**Wilmington Airport**” has the **highest average departure delay** among the top 15 airports analyzed.
- “**Spirit Airlines**” shows the **largest number of delays** due to **air system issues**.

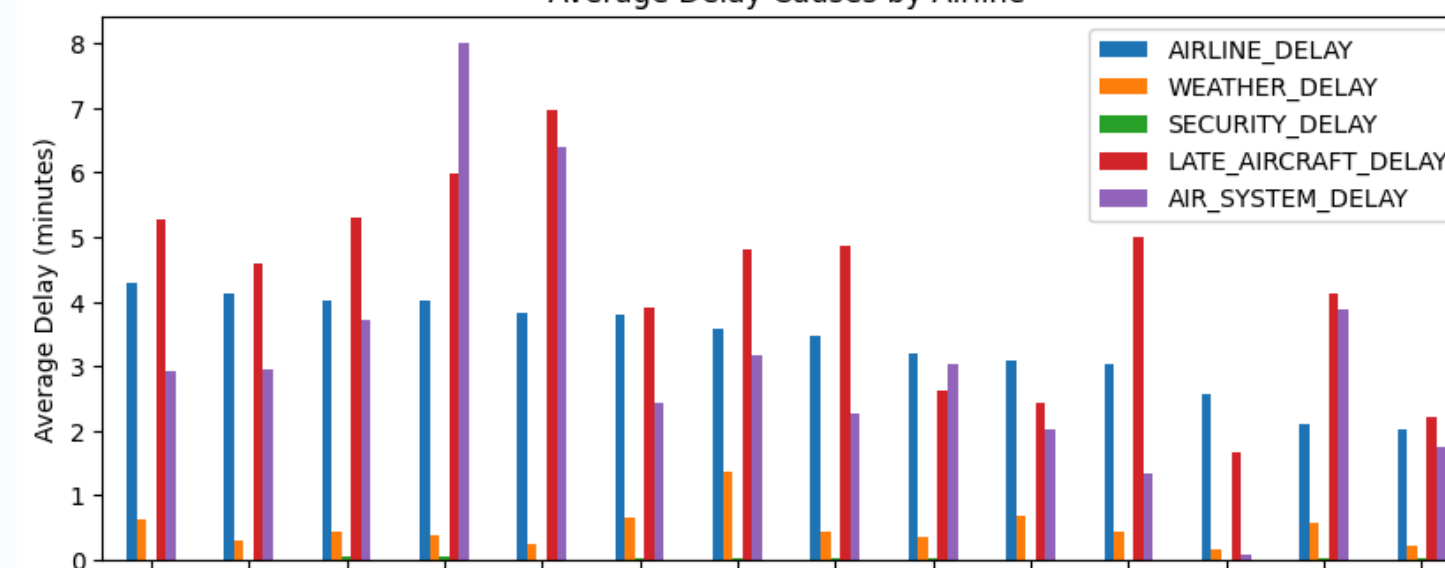
Total Contribution of Delay Causes



Top 15 Airports with Highest Average Departure Delay



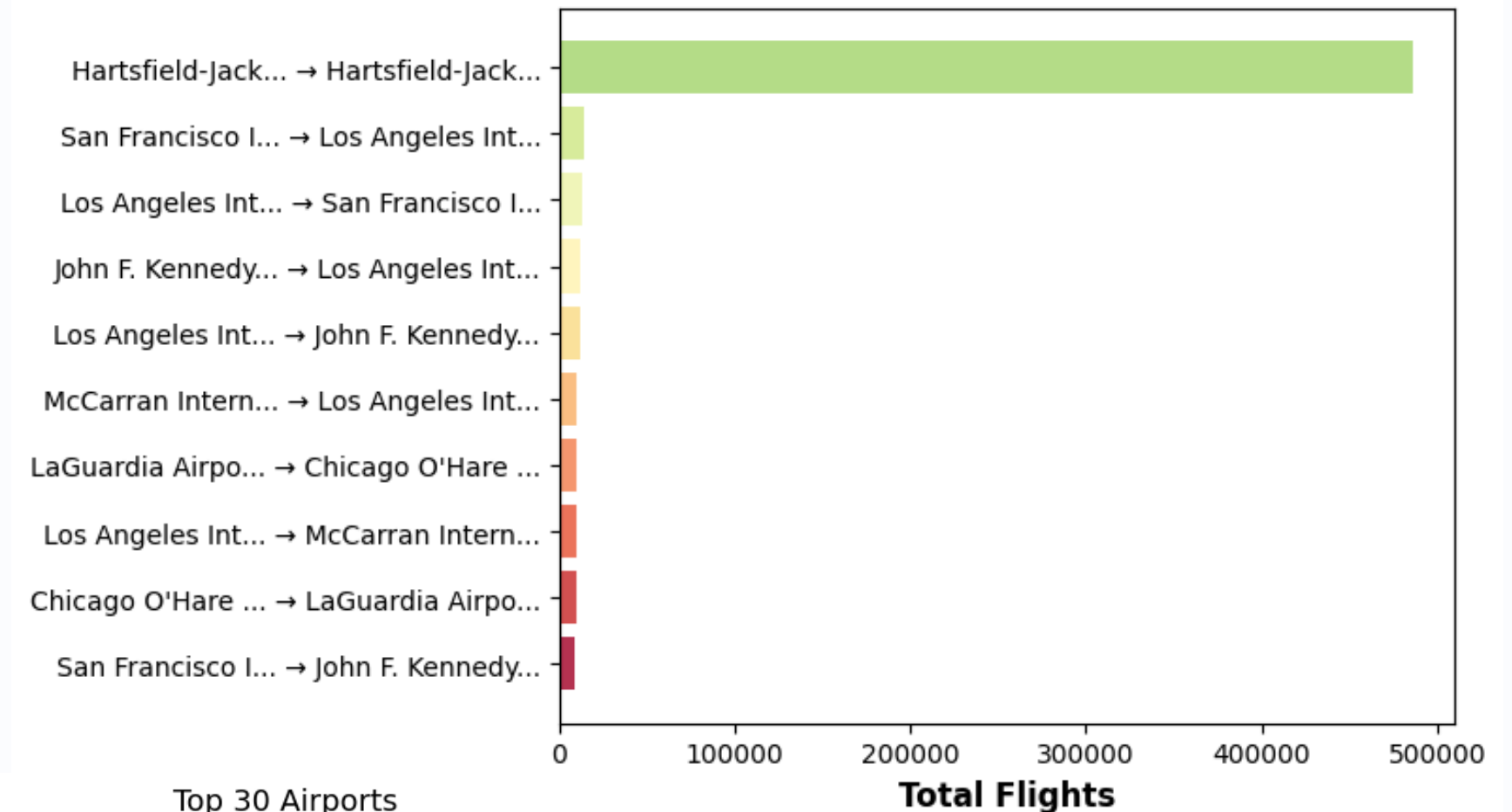
Average Delay Causes by Airline



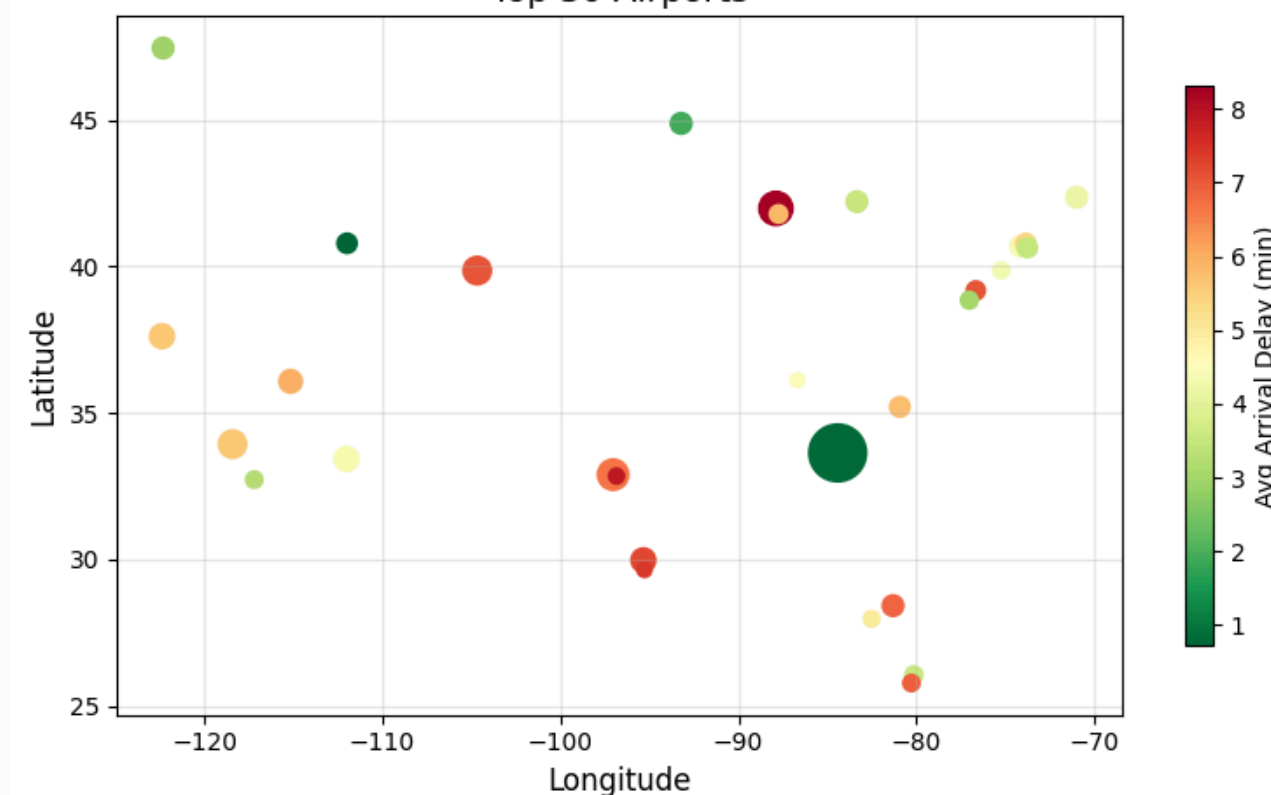
AIRPORT & ROUTE-LEVEL ANALYSIS

- Major hub airports handle more flights and generally show higher average arrival delays.
- Delays vary by location, with some busy airports still maintaining lower delays.
- A few routes dominate total flight volume, especially between major hub airports

Top 10 Busiest Routes by Flight Volume

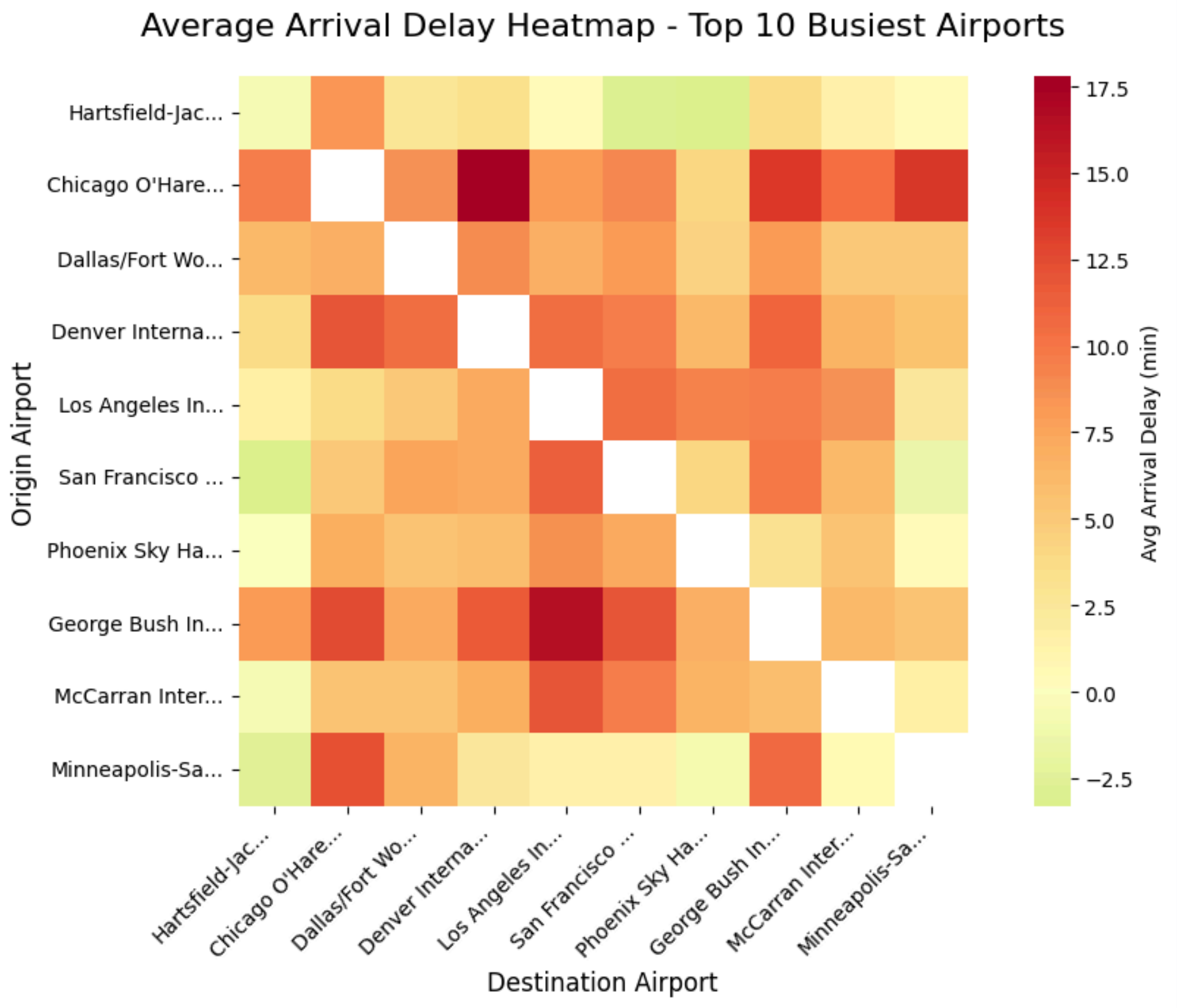


Top 30 Airports



DELAY HEATMAP

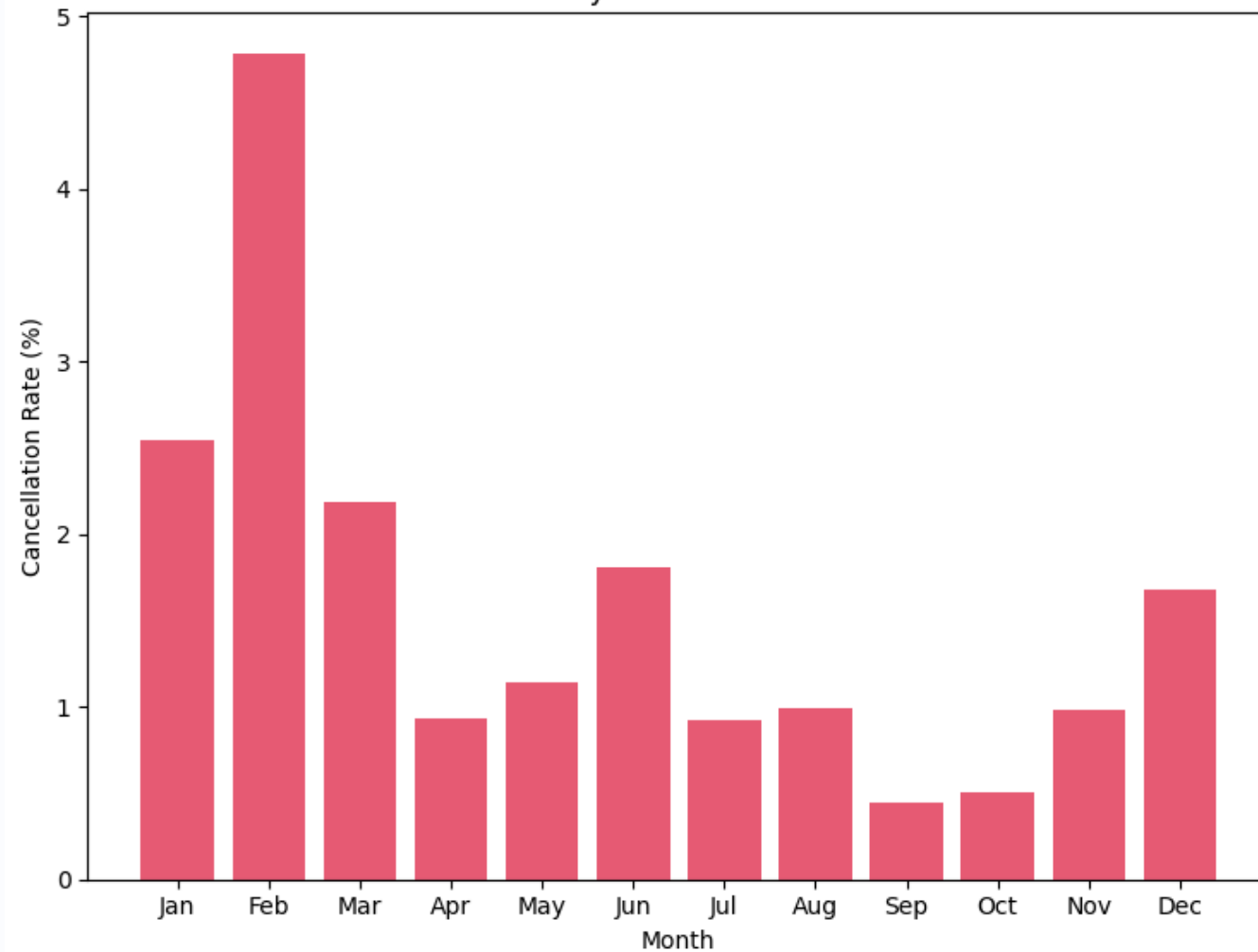
Average arrival delays vary significantly across airport pairs, with major hub-to-hub routes showing consistently higher delays.



SEASONAL & CANCELLATION ANALYSIS

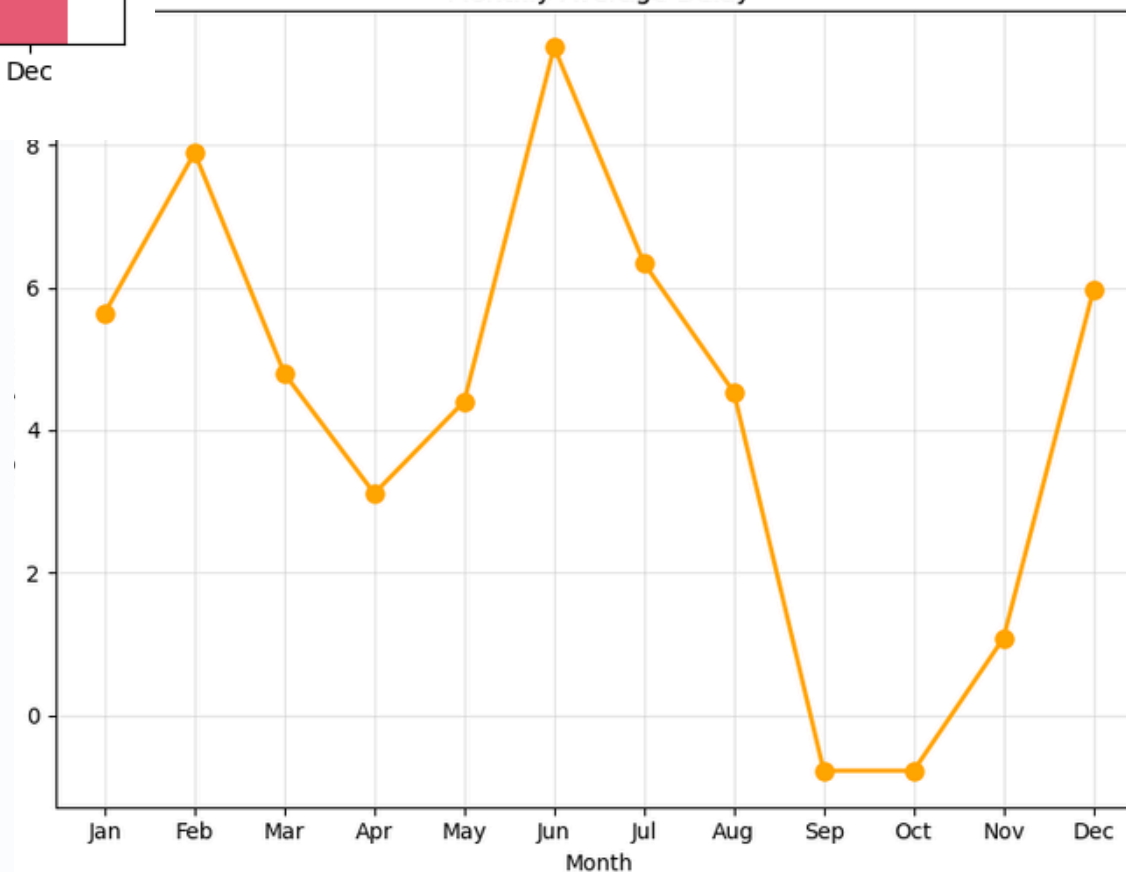


Monthly Cancellation Rate

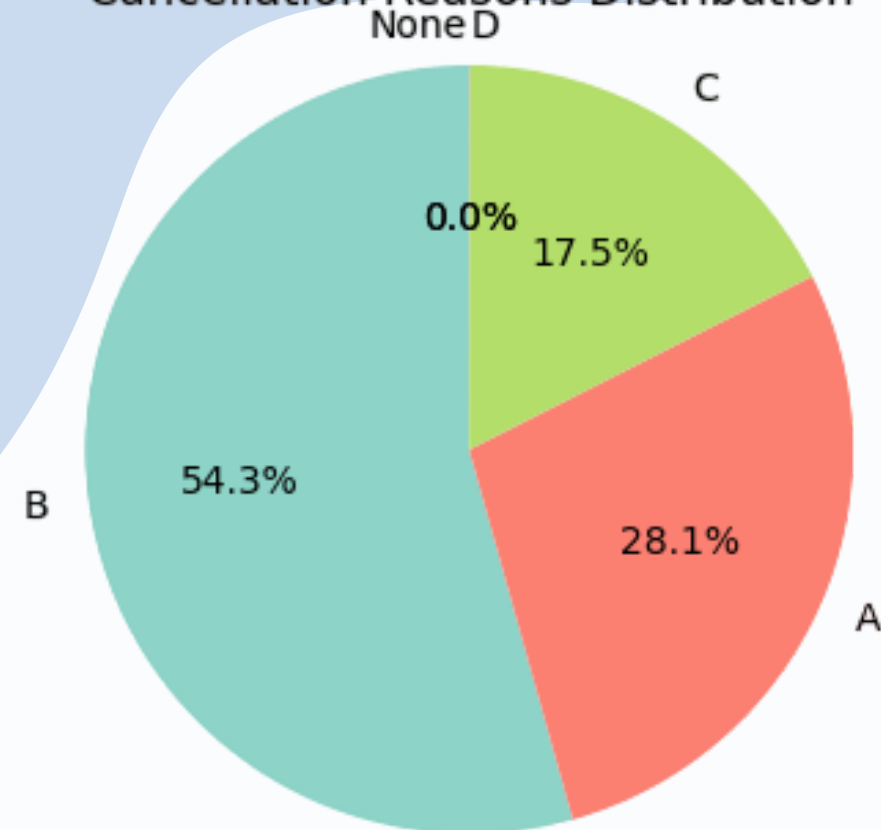


- Observed **Highest Cancellations in February**, **lowest** during **September–October**, showing fewer disruptions in early fall.
- **Average delays** peak around **June–July** and drop to the lowest levels in **September–October**.

Monthly Average Delay

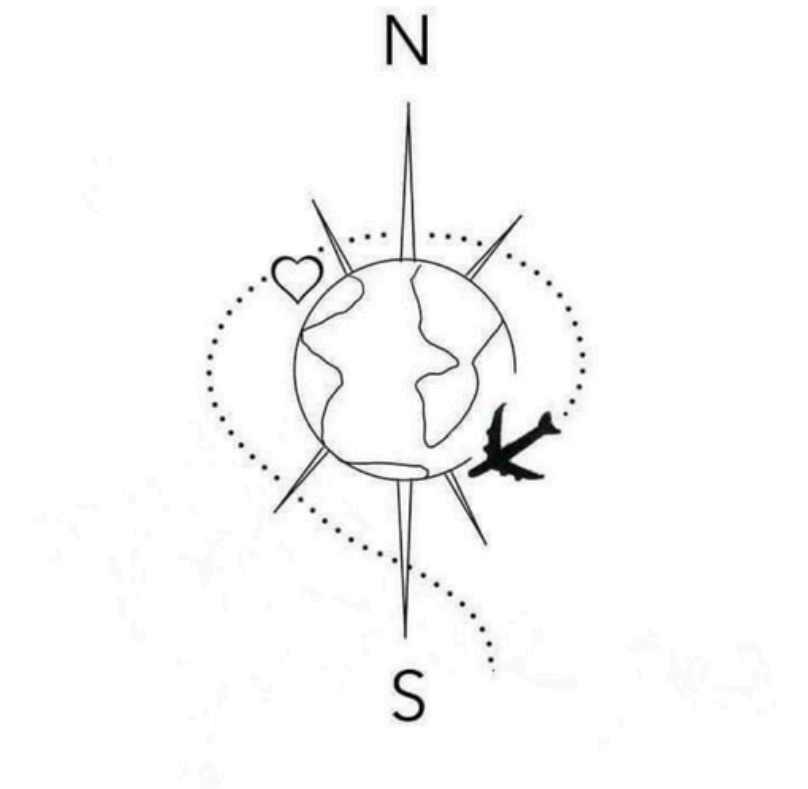


Cancellation Reasons Distribution



Seasonal Insights:

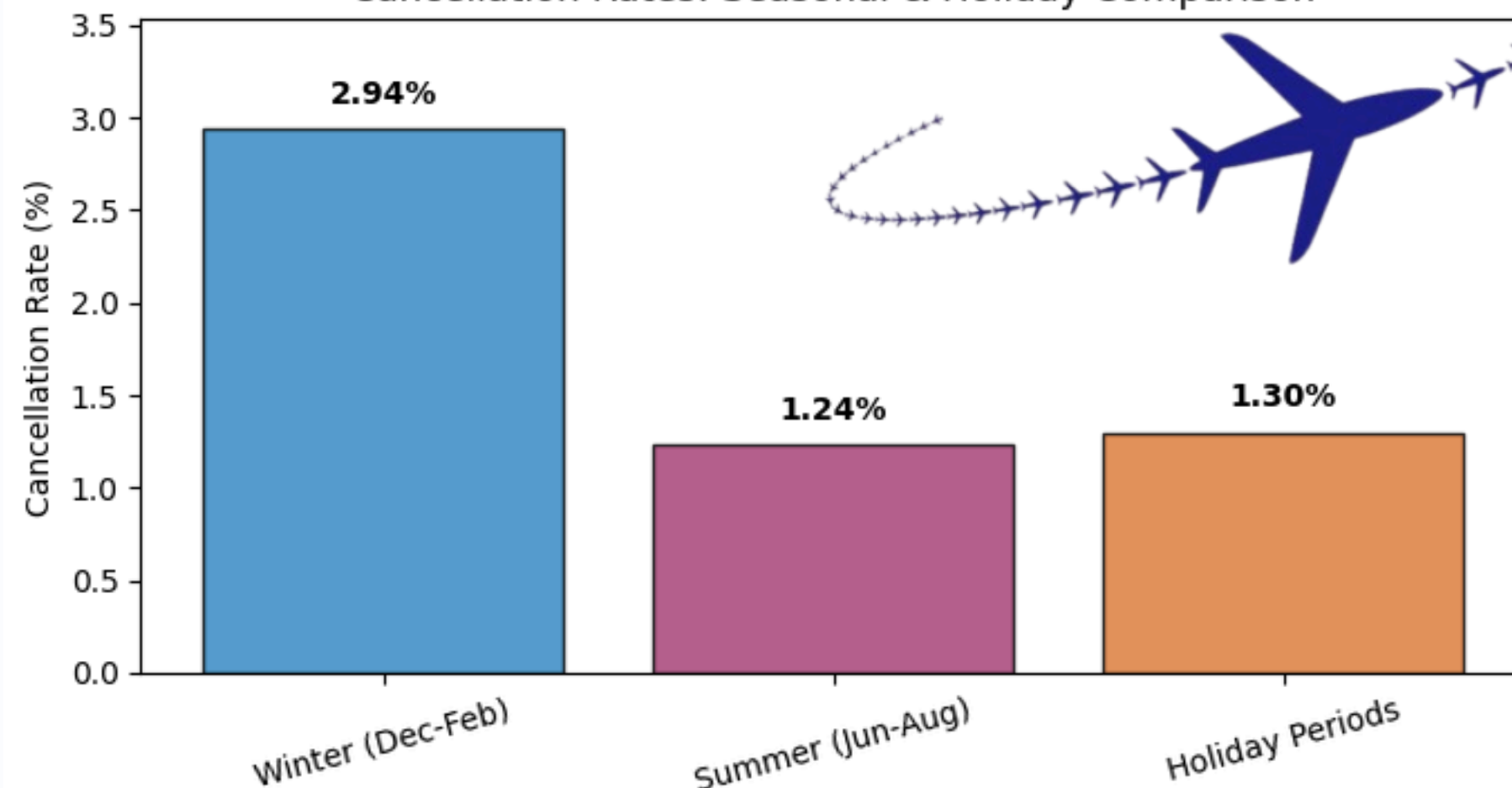
- Winter Cancellation Rate: 2.94%
- Summer Cancellation Rate: 1.24%
- Holiday Period Rate: 1.30%



Cancellation Reasons:

- Weather: 48851 flights (54.3%)
- Carrier: 25262 flights (28.1%)
- NAS: 15749 flights (17.5%)
- Security: 22 flights (0.0%)
- None: 0 flights (0.0%)

Cancellation Rates: Seasonal & Holiday Comparison



CORE FINDINGS IN AIRLINE NETWORK

- Delays are not random — they **accumulate as the day progresses**, especially on busy workdays.
- High traffic alone doesn't cause delays — **operational efficiency at hubs** determines outcomes.
- Most delays are **self-propagating**, caused by earlier delays rolling forward through the network.
- Seasonal reliability is driven more by **operational conditions** than by passenger volume.
- Weather is the main reason flights are cancelled — but **not the main reason flights are delayed**.





THANK
YOU

